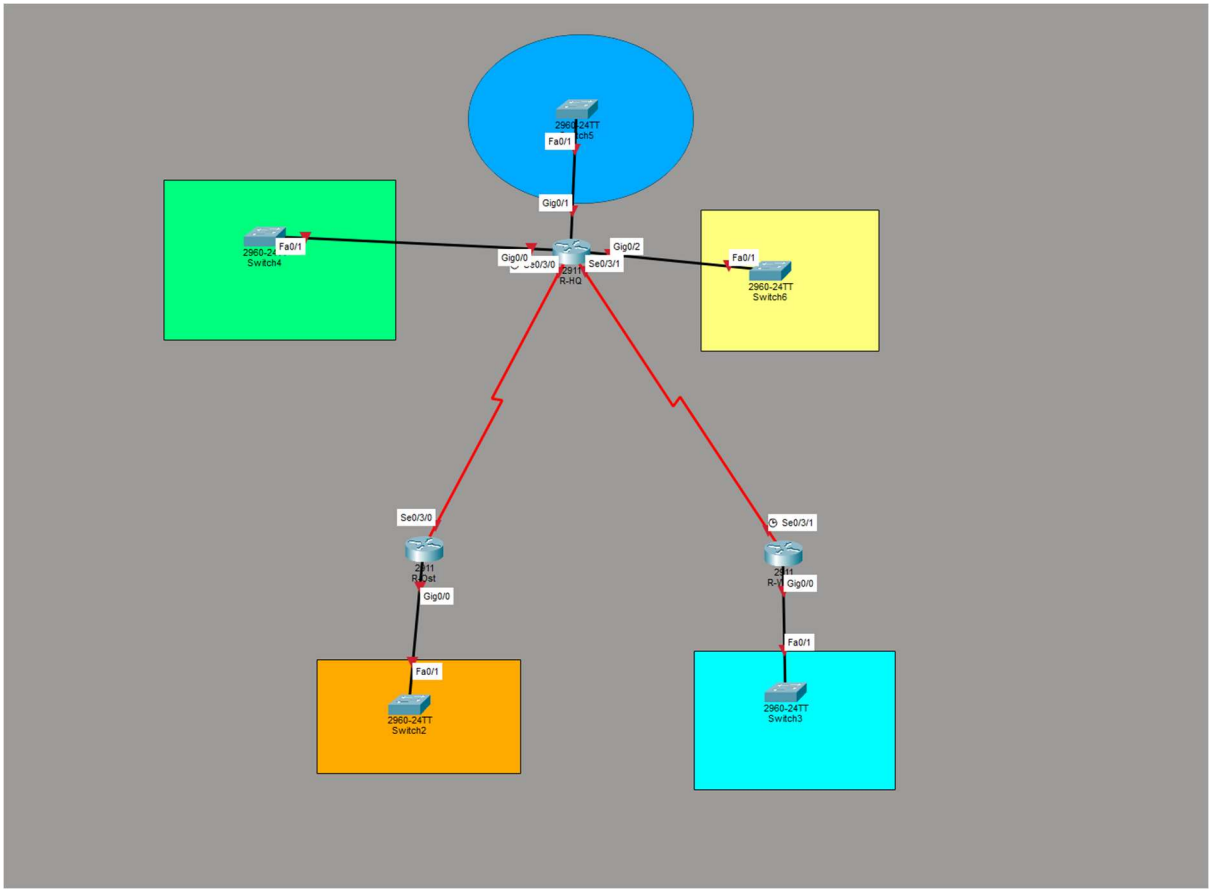
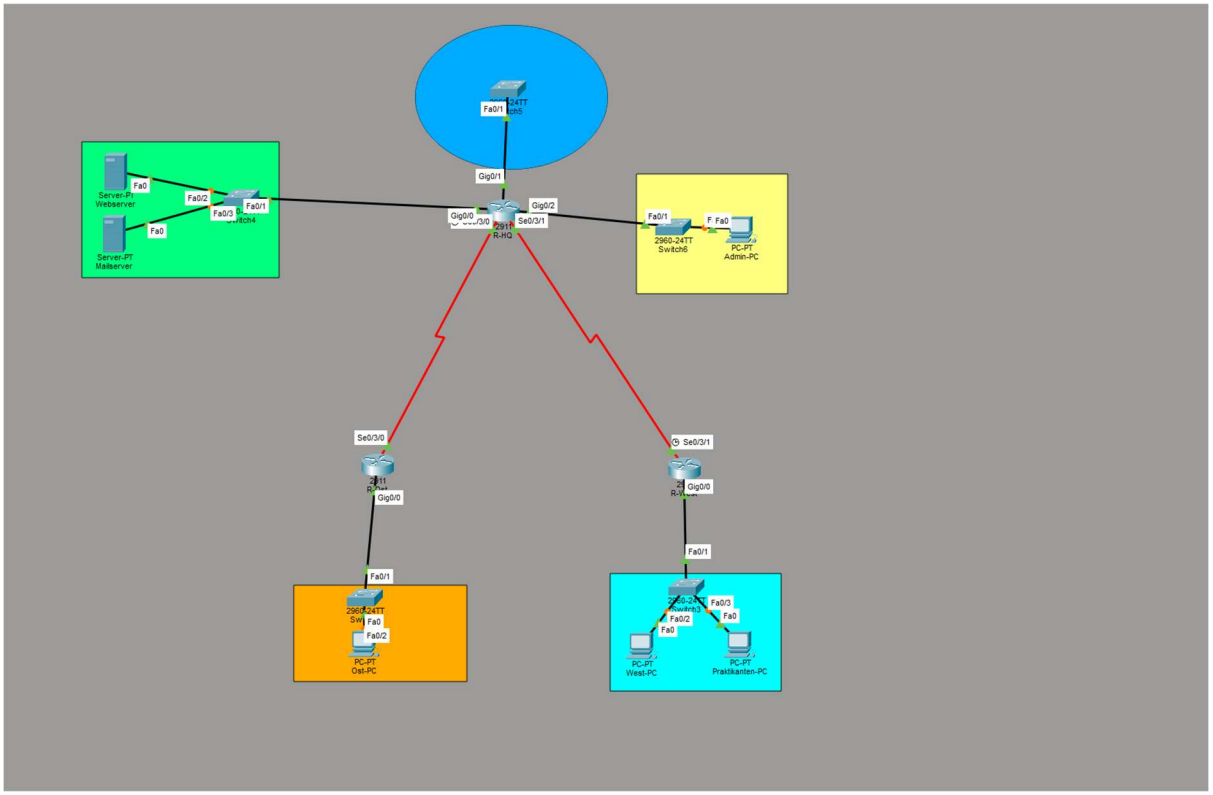

Fa0/1
Gig0/1
Gig0/0
Se0/3/0
Se0/3/1
Gig0/2
Fa0/1
Switch4
R-HQ
Switch6
Se0/3/0
Se0/3/1
Gig0/0
Fa0/1
Switch2
Switch3


Fa0/1
Gig0/1
Gig0/0
Se0/3/0
Se0/3/1
Gig0/2
Fa0
Server-PT
Webserver
Server-PT
Mailserver
Switch4
R-HQ
Switch6
PC-PT
Admin-PC
Gig0/0
Fa0/1
Fa0/2
Ost-PC
West-PC
Praktikanten-PC

```
R-HQ#
R-HQ#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     172.20.5.1      YES manual up                    up
GigabitEthernet0/1     203.0.113.6     YES manual up                    up
GigabitEthernet0/2     172.20.5.30     YES manual up                    up
Serial0/3/0            172.20.9.2      YES manual up                    up
Serial0/3/1            172.20.9.6      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R-HQ#
```

```
R-HQ#show access-lists
Extended IP access list WEBONLY
    10 permit tcp 172.20.4.0 0.0.0.191 host 172.20.5.14 eq www
    20 permit tcp 172.20.4.0 0.0.0.191 host 172.20.5.14 eq 443
    30 deny ip 172.20.4.0 0.0.0.191 172.20.5.0 0.0.0.15
    40 permit ip any any
Standard IP access list 11
    10 permit 172.20.5.16 0.0.0.15
```

R-HQ#
R-HQ#show ip interface
GigabitEthernet0/0 is up, line protocol is up (connected)
Internet address is 172.20.5.1/28
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is WEBONLY
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachables are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled

Input features: MCI Check
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled
GigabitEthernet0/1 is up, line protocol is up (connected)
Internet address is 203.0.113.6/29
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachables are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled
Input features: MCI Check
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled
GigabitEthernet0/2 is up, line protocol is up (connected)
Internet address is 172.20.5.30/28
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set

Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachables are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
BGP Policy Mapping is disabled
Input features: MCI Check
WCCP Redirect outbound is disabled
WCCP Redirect inbound is disabled
WCCP Redirect exclude is disabled
Serial0/3/0 is up, line protocol is up (connected)
Internet address is 172.20.9.2/30
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachables are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled

IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
WCCP Redirect outbound is disabled
WCCP Redirect exclude is disabled
BGP Policy Mapping is disabled
Serial0/3/1 is up, line protocol is up (connected)
Internet address is 172.20.9.6/30
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500
Helper address is not set
Directed broadcast forwarding is disabled
Outgoing access list is not set
Inbound access list is not set
Proxy ARP is enabled
Security level is default
Split horizon is enabled
ICMP redirects are always sent
ICMP unreachables are always sent
ICMP mask replies are never sent
IP fast switching is disabled
IP fast switching on the same interface is disabled
IP Flow switching is disabled
IP Fast switching turbo vector
IP multicast fast switching is disabled
IP multicast distributed fast switching is disabled
Router Discovery is disabled
IP output packet accounting is disabled
IP access violation accounting is disabled
TCP/IP header compression is disabled
RTP/IP header compression is disabled
Probe proxy name replies are disabled
Policy routing is disabled
Network address translation is disabled
WCCP Redirect outbound is disabled
WCCP Redirect exclude is disabled
BGP Policy Mapping is disabled
Vlan1 is administratively down, line protocol is down
Internet protocol processing disabled

R-HQ#

```
R-HQ#show access-lists WEBONLY
Extended IP access list WEBONLY
    deny ip host 172.20.4.150 host 172.20.5.13
    permit tcp 172.20.4.0 0.0.0.191 host 172.20.5.14 eq www
    permit tcp 172.20.4.0 0.0.0.191 host 172.20.5.14 eq 443
    deny ip 172.20.4.0 0.0.0.191 172.20.5.0 0.0.0.15
    permit ip any any

R-HQ#
R-HQ#
```

```
R-Ost>enable
R-Ost#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R-Ost(config)#access-list 10 deny 172.20.4.128 0.0.0.63
R-Ost(config)#access-list 10 permit any
R-Ost(config)#interface g0/0
R-Ost(config-if)# ip access-group 10 out
R-Ost(config-if)#end
R-Ost#show access-lists
Standard IP access list 10
    10 deny 172.20.4.128 0.0.0.63
    20 permit any (40 match(es))

R-Ost#copy run start
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
R-Ost#
R-Ost#show access-lists
Standard IP access list 10
    10 deny 172.20.4.128 0.0.0.63
    20 permit any (40 match(es))

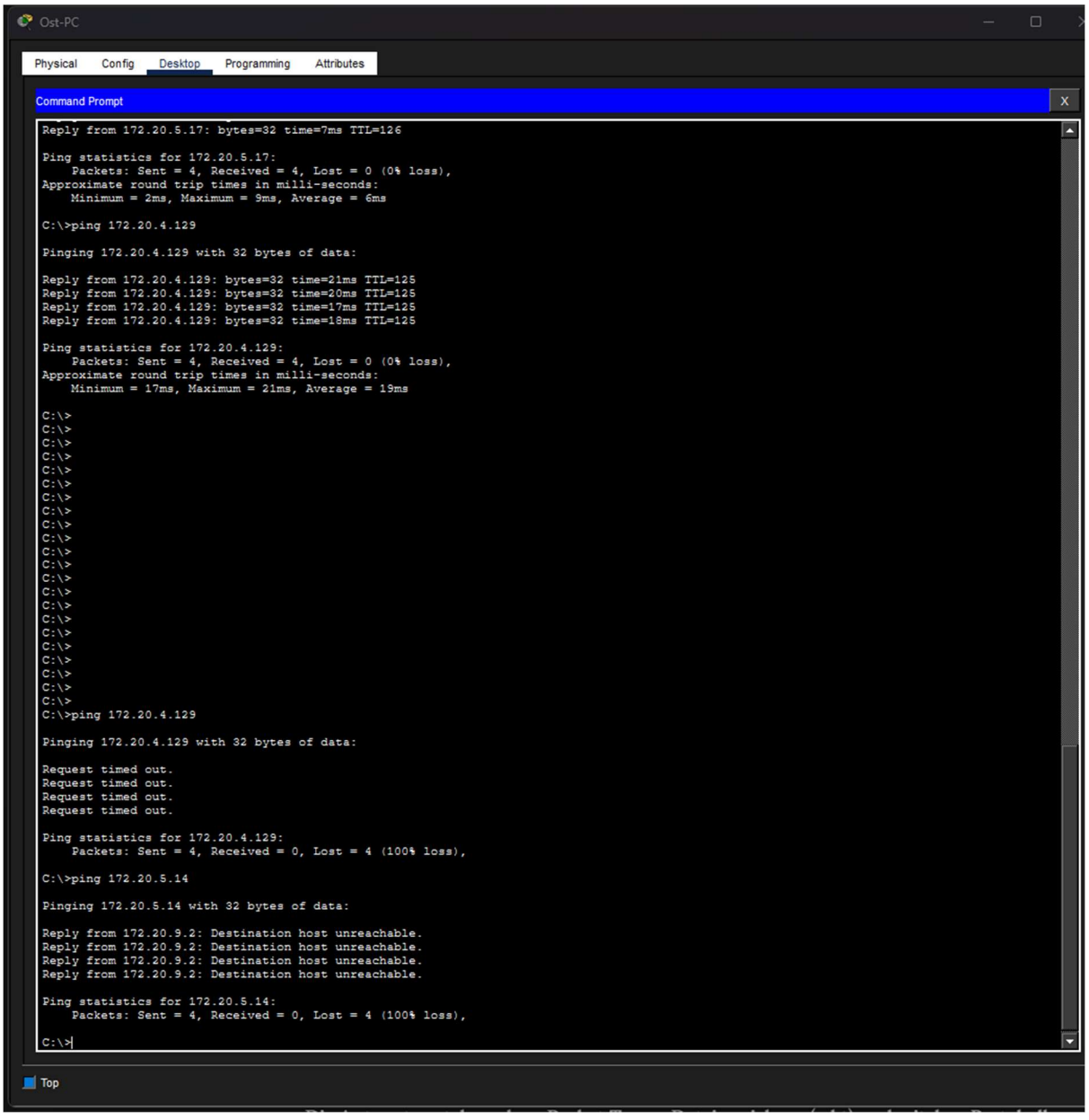
R-Ost#
```

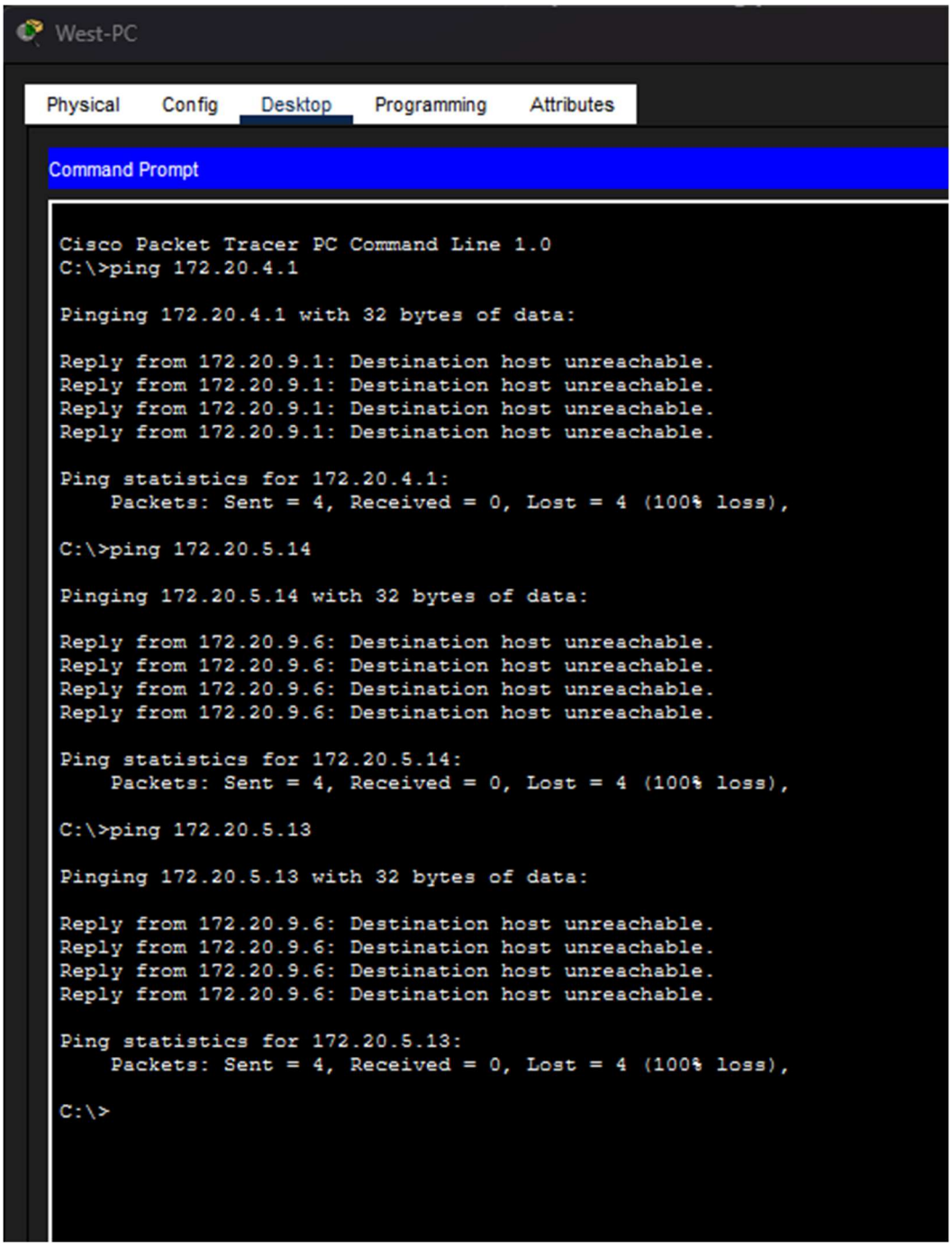
```
R-HQ#
R-HQ#show access-lists
Extended IP access list WEBONLY
    5 deny ip host 172.20.4.150 host 172.20.5.13
    10 permit tcp 172.20.4.0 0.0.0.191 host 172.20.5.14 eq www
    20 permit tcp 172.20.4.0 0.0.0.191 host 172.20.5.14 eq 443
    30 deny ip 172.20.4.0 0.0.0.191 172.20.5.0 0.0.0.15
    40 permit ip any any
Standard IP access list 11
    10 permit 172.20.5.16 0.0.0.15

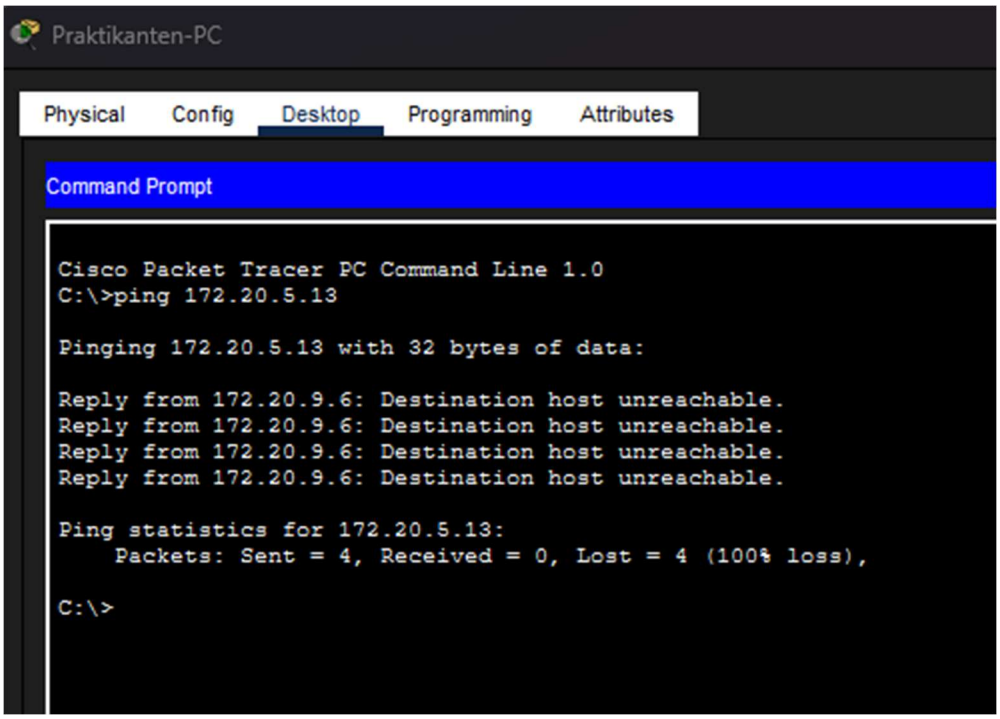
R-HQ#show ip interface g0/0 | include access list
  Outgoing access list is WEBONLY
  Inbound  access list is not set
R-HQ#show running-config | section line vty
line vty 0 4
 access-class 11 in
 login local
 transport input ssh
line vty 5 15
 access-class 11 in
 login local
 transport input ssh
R-HQ#
R-HQ#
R-HQ#
```

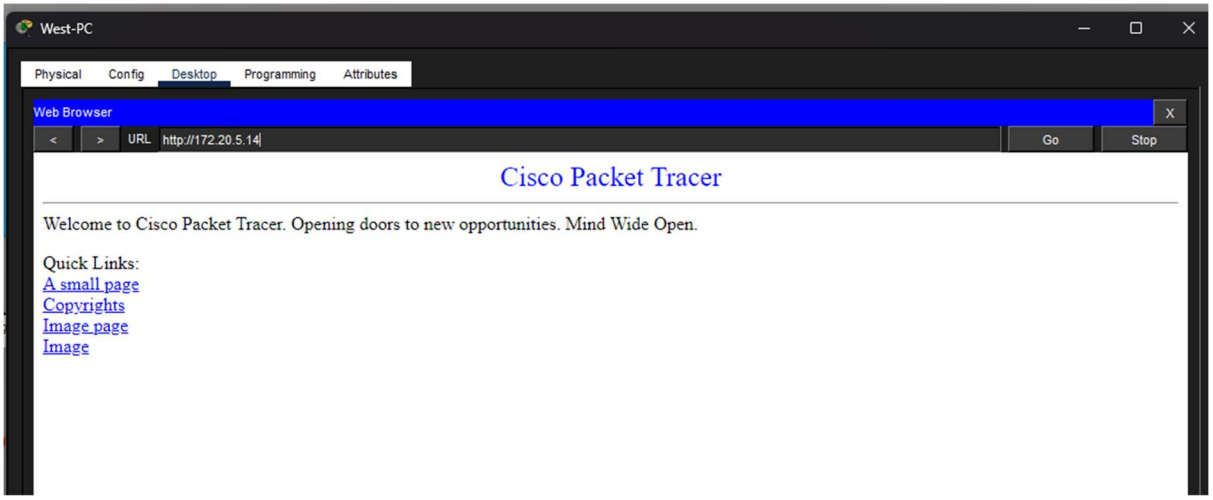
```
R-Ost#
R-Ost#show access-lists
Standard IP access list 10
    10 deny 172.20.4.128 0.0.0.63
    20 permit any

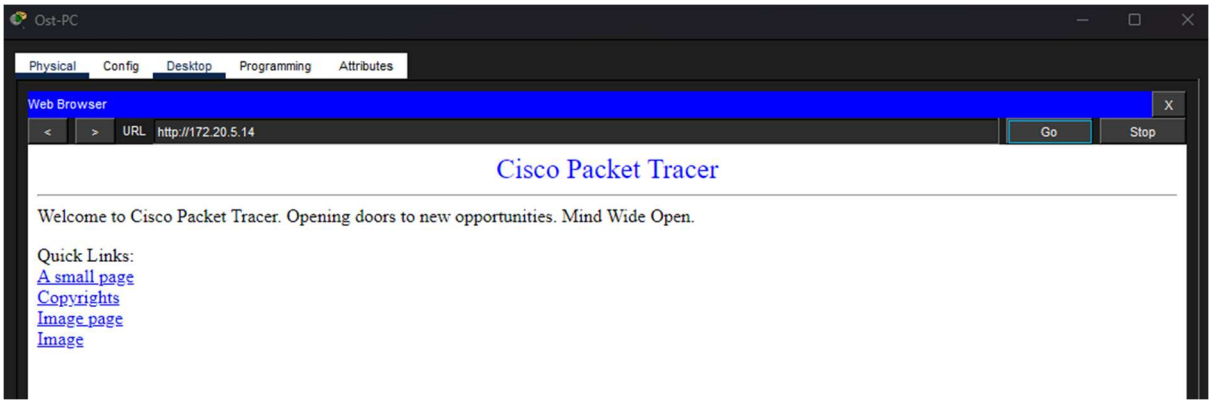
R-Ost#show ip interface g0/0 | include access list
  Outgoing access list is 10
  Inbound  access list is 10
R-Ost#
R-Ost#
```

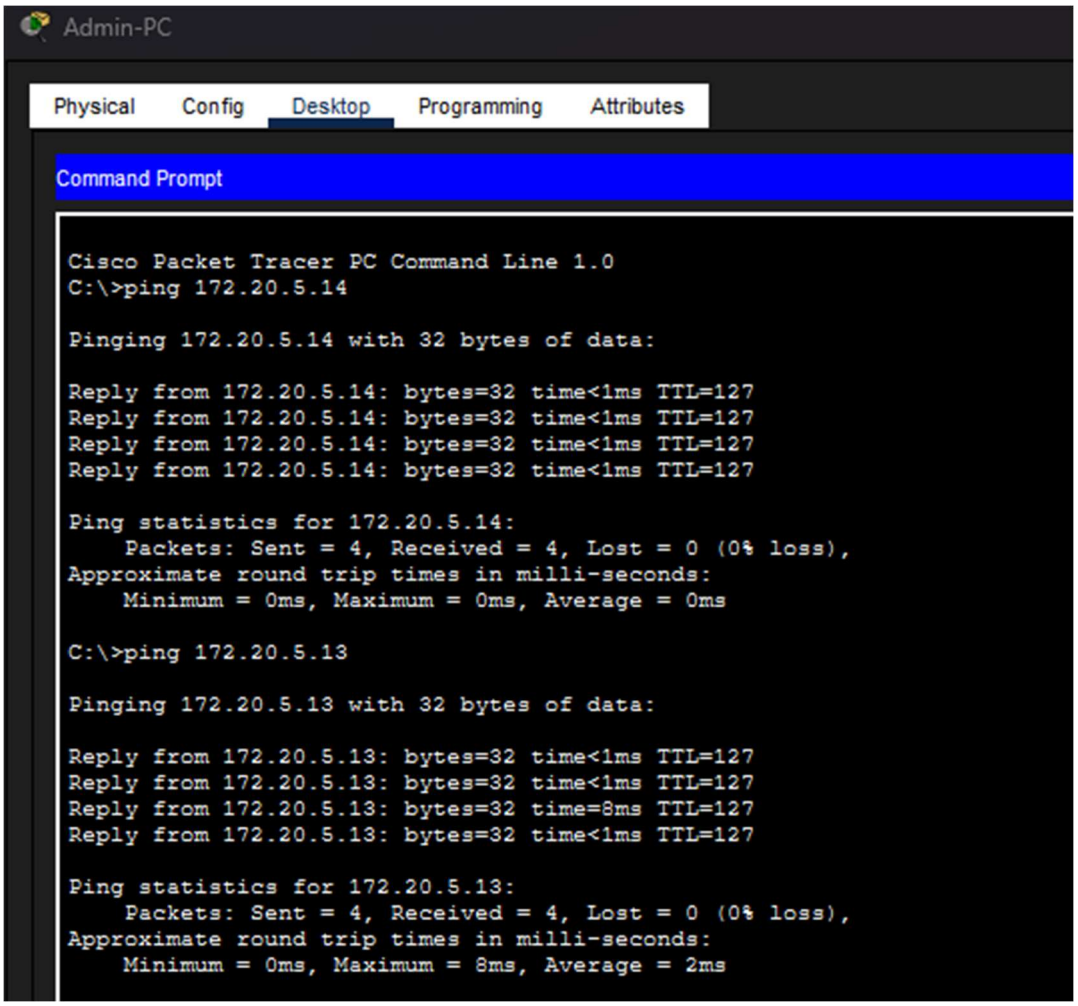


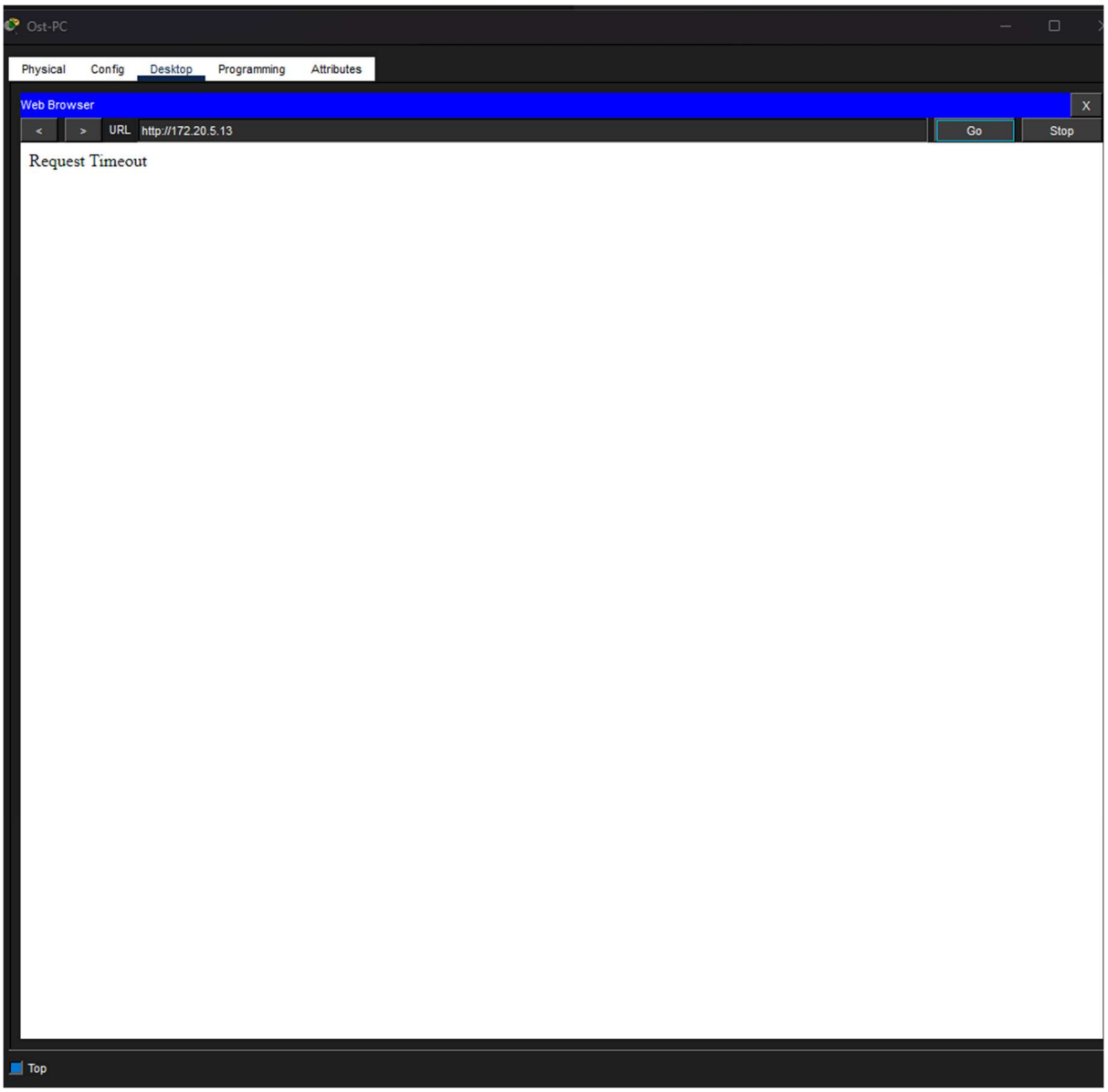

West-PC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.20.4.1
Pinging 172.20.4.1 with 32 bytes of data:
Reply from 172.20.9.1: Destination host unreachable.
Reply from 172.20.9.1: Destination host unreachable.
Reply from 172.20.9.1: Destination host unreachable.
Reply from 172.20.9.1: Destination host unreachable.
Ping statistics for 172.20.4.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 172.20.5.14
Pinging 172.20.5.14 with 32 bytes of data:
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Ping statistics for 172.20.5.14:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 172.20.5.13
Pinging 172.20.5.13 with 32 bytes of data:
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Ping statistics for 172.20.5.13:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>


Praktikanten-PC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.20.5.13
Pinging 172.20.5.13 with 32 bytes of data:
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Reply from 172.20.9.6: Destination host unreachable.
Ping statistics for 172.20.5.13:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>


West-PC
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL
http://172.20.5.14
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image


Ost-PC
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL
http://172.20.5.14
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image


Admin-PC
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.20.5.14
Pinging 172.20.5.14 with 32 bytes of data:
Reply from 172.20.5.14: bytes=32 time<1ms TTL=127
Reply from 172.20.5.14: bytes=32 time<1ms TTL=127
Reply from 172.20.5.14: bytes=32 time<1ms TTL=127
Reply from 172.20.5.14: bytes=32 time<1ms TTL=127
Ping statistics for 172.20.5.14:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 172.20.5.13
Pinging 172.20.5.13 with 32 bytes of data:
Reply from 172.20.5.13: bytes=32 time<1ms TTL=127
Reply from 172.20.5.13: bytes=32 time<1ms TTL=127
Reply from 172.20.5.13: bytes=32 time=8ms TTL=127
Reply from 172.20.5.13: bytes=32 time<1ms TTL=127
Ping statistics for 172.20.5.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 8ms, Average = 2ms




```
C:\>
C:\>ssh -l admin 172.20.5.30

Password:



R-HQ>show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R-HQ>
```

```
C:\>
C:\>ssh -l admin 172.20.5.30

% Connection refused by remote host
C:\>
```